GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M4: Traffic and AI

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1. Purpose

M4 brings the city to life. It finds routes and travel times on the road network. It moves the NPC cars that make the traffic. It also drives the AI bots that fill the empty places in a game. A bot makes the same control message that a phone sends, so for M1 a bot and a human player look the same.

Runs on	Main computer inside the game server; M1 runs it once in every tick
Technology	TypeScript; our own shortest-path search (Dijkstra); a car-following model for NPC cars; a state machine and path following for bots; seeded random numbers
Requirements	R1 R2 R4 R6 R11

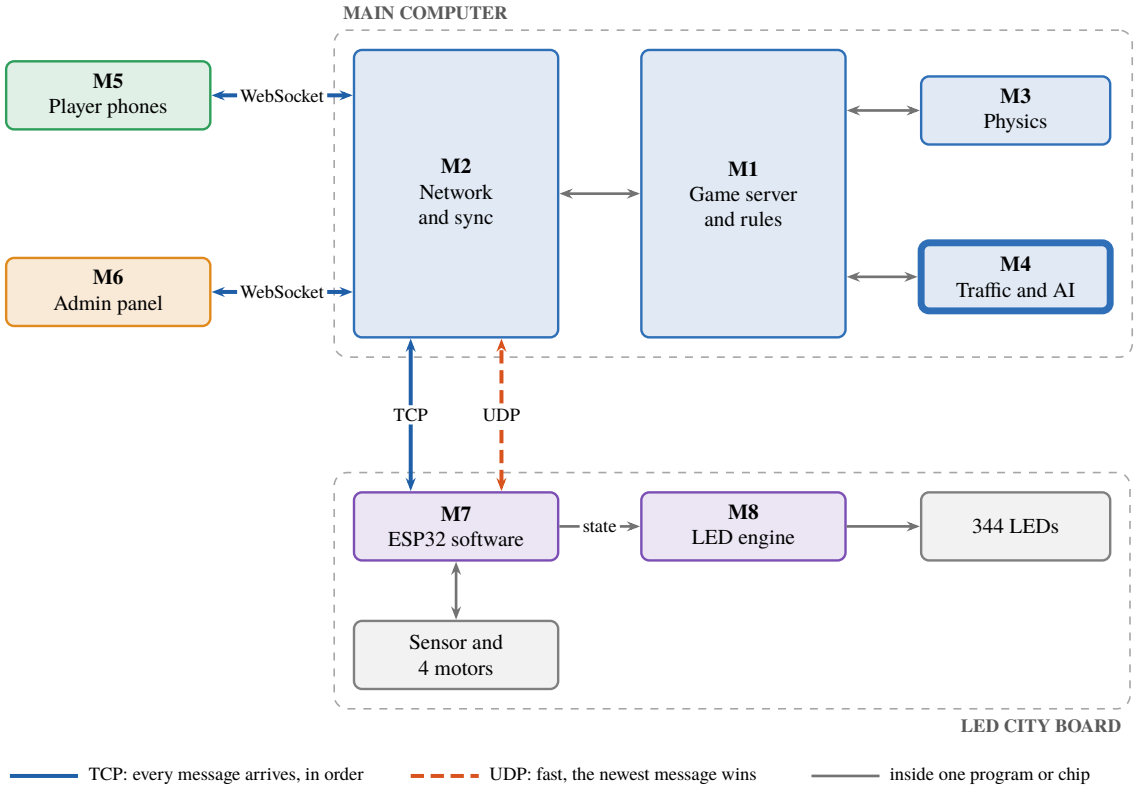


Figure 1: Where M4 sits in the system. M4 talks only to M1, inside the same program.

2. Requirements covered

ID	Requirement	What M4 does for it
R1	Multiplayer game	Bots take the empty places and play like players
R2	Delivery cycle	Travel-time table from each store, which M1 uses for fair near and far points
R4	Traffic	NPC cars make real traffic on the roads
R6	Obstacles	Heavy traffic adds slow cars; an accident closes a lane; road work closes the road for routes and cars

ID	Requirement	What M4 does for it
R11	Independent development	Talks only to M1 with plain data; tested with its own test program

3. Inputs and outputs

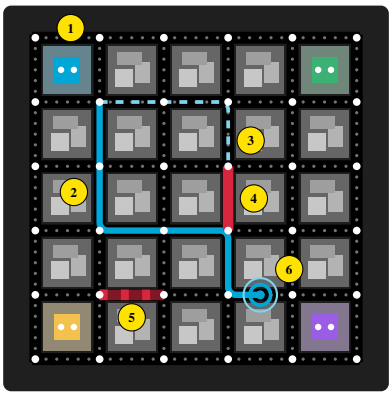
Dir.	Data	From / to	Content
In	City plan	Shared file	Intersections, roads, lanes, stores, possible delivery points
In	Game data	M1	Every tick: players, obstacles with a version number, car-hit events
In	Parameters	M6, through M1	Number of NPC cars, traffic effects, bot skill
Out	NPC cars	M1, then M3, M5, M6	Position, direction, speed, lane, size
Out	Bot controls	M1, then M3	The same record as a phone: throttle, brake, steer, deliver
Out	Travel times	M1	Seconds from each team store to every possible delivery point
Out	Traffic metrics	M1, then M6	Number of NPC cars, mean NPC speed, bot route changes

4. Main components

Component	What it does	Related modules
Road graph	36 intersections and 60 two-way roads with lanes and turn costs	—
Route planner	Fastest routes and the travel-time table; computed again only when the obstacles change	M1
NPC traffic	Moves, adds and removes NPC cars	M1, M3
Bot drivers	One state machine for each bot	M1, M3
Traffic metrics	Values for the admin graphs	M6

5. How it works

The road graph and routes. The city has 6 roads in each direction. Where two roads cross, there is an intersection, so the graph has 36 intersections. Between them there are 60 road segments, and cars can drive both ways. Each segment has a cost: the expected time to drive it. Turning left or making a U-turn adds a small extra cost. Obstacles change the costs. Heavy traffic and an accident make a segment slower, and road work closes it. M4 uses Dijkstra's shortest-path search to find the fastest routes. The graph is small, so one search takes much less than a millisecond. M4 computes the routes again only when the version number of the obstacle list changes. From the results it makes the travel-time table: the time from each of the four stores to every possible delivery point. M1 uses this table to choose fair near and far points.



1 Start: the team store
2 Fastest route, found by Dijkstra
3 Shorter road, but closed
4 Road work: cost is infinite
5 Heavy traffic: cost is higher
6 Delivery point
◦ Intersection, a node of the graph

Figure 2: A route on the road graph, drawn on the board. Obstacles change the cost of the roads.

NPC traffic. Each NPC car follows a lane and picks a new turn at each intersection. A car keeps a safe gap behind the car or the motorcycle in front of it. If the gap gets small, it slows down smoothly, and it speeds up again when the road is free. A car enters an intersection only when the intersection is free, so cars do not drive through each other. New cars never appear close to a player. If a car is stuck for 10 seconds, M4 removes it and adds it again somewhere else, so the traffic never locks up. The admin sets the number of cars, and heavy-traffic obstacles add extra slow cars on their road.

Bots. Every bot has a small state machine that follows the delivery cycle, as the figure below shows. The bot drives to the store, gets the package, drives to the delivery point, slows down and presses "Deliver", and then goes back. On the way it follows the route from the planner. It steers toward a point a little ahead on the path, so the motion looks smooth. If it falls, it waits for the penalty time. If it is stuck, it asks for a new route. One "bot skill" value sets the top speed, the reaction time and some steering noise, so the admin can make bots weaker or stronger.

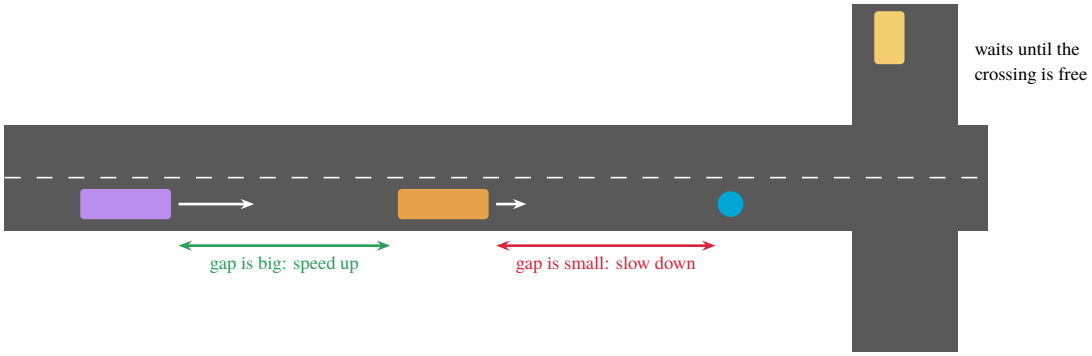


Figure 3: How NPC cars drive: a safe gap behind the next vehicle, and a free crossing before they enter.

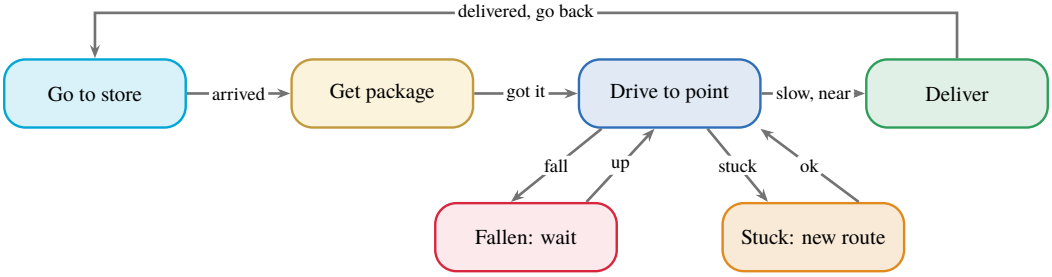


Figure 4: The state machine of one bot.

6. Testing and demonstration

A small test program plays the role of M1 and draws the city from above. We test routes with and without closed roads, equal travel times for all teams in a city that is the same on all sides, 80 cars for 30 minutes without a traffic lock, obstacles that change during play, 8 bots that finish deliveries, and the same result with the same seed. In the demonstration the admin adds obstacles and more traffic, and the cars and bots change their way at once.